Java Mastery: Data, Web & Backend

Tagline: *Go from zero to deploying interactive data web apps in 8 weeks—using purely Java.*

Here is your detailed, 8-week, 5-days-a-week curriculum.

PHASE 1: CORE JAVA MASTERY (WEEKS 1–5)

JAVA MASTERY (WEEKS 1–8)

This 8-week roadmap is designed to make you strong in Core Java, Advanced Java, Database, and Frameworks for interviews and real-time projects.

WEEK 1: JAVA FUNDAMENTALS

• **Day 1: Introduction to Java:** History of Java, features of Java, applications of Java, edition types (**SE, EE, ME**).

• **Day 2: Java Environment Setup:** Install JDK, understand JRE & JVM, configure IDE (**IntelliJ / Eclipse / VS Code**), write first Java program.

• **Day 3: Variables & Data Types:** Primitive and non-primitive data types, variables, literals, type casting.

• **Day 4: Operators & Input/Output:** Arithmetic, relational, logical, bitwise, unary, ternary operators. Scanner class for user input.

• **Day 5: Control Statements:** if, if-else, nested if, else-if ladder, switch statement.

o **Lab:** Build simple calculator and student grade calculator.

WEEK 2: LOOPS & ARRAYS

• **Day 1: Loops Basics:** for, while, do-while loops.

• **Day 2: Jump Statements & Patterns:** break, continue, nested loops, pattern programs.

• **Day 3: Arrays Basics:** 1D arrays, array declaration, initialization, traversing arrays.

• **Day 4: Advanced Arrays:** 2D arrays, jagged arrays, multidimensional arrays.

• **Day 5: Searching & Sorting:** Linear search, binary search, bubble sort, selection sort.

o **Lab:** Build array-based employee record system.

WEEK 3: STRINGS & METHODS

• **Day 1: String Basics:** Immutable strings, string pool, string methods.

• **Day 2: Mutable Strings:** StringBuffer, StringBuilder, mutable string operations.

• **Day 3: Methods in Java:** Method declaration, method calling, parameters, return types.

• **Day 4: Advanced Methods:** Method overloading, recursion, factorial, Fibonacci.

• **Day 5: String Interview Programs:** Reverse string, palindrome, count characters, duplicate characters.

o **Lab:** Build text analyser application.

WEEK 4: OOP CONCEPTS

• **Day 1: Class & Object:** Creating classes and objects, object memory allocation.

• **Day 2: Constructor & Keywords:** Constructor types, this, static, final.

• **Day 3: Inheritance:** Single, multilevel, hierarchical inheritance.

• **Day 4: Polymorphism:** Compile-time and runtime polymorphism.

• **Day 5: Encapsulation, Abstraction & Interface:** Getter/setter methods, abstract class, interfaces.

o **Lab:** Build banking system mini project.

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WEEK 5: EXCEPTION HANDLING & COLLECTIONS

• **Day 1: Exception Handling Basics:**
Exception hierarchy, checked vs unchecked exceptions, try, catch, finally, multiple catch blocks.

• **Day 2: Advanced Exception Handling:**
throw, throws, custom exceptions, user-defined exceptions, best practices.

• **Day 3: Collections Framework Overview:**
Introduction to Collection Framework, differences between List, Set, Queue, and Map.

• **Day 4: List & Set Implementations:**
ArrayList, LinkedList, Vector, Stack, HashSet, Linked HashSet, Tree Set.

• **Day 5: Queue & Map Implementations:**
Priority Queue, Deque, Array Deque, HashMap, Linked HashMap, Tree Map, Hash table, Iterator, List Iterator.

o **Lab:** Build **Student Record Management System** using Collections.

PHASE 2: CORE JAVA MASTERY (WEEKS 6–8)

WEEK 6: ADVANCED JAVA

• **Day 1: Multithreading:**
Thread creation using Thread class and Runnable interface, thread lifecycle, thread priorities.

• **Day 2: Synchronization & Inter-thread Communication:**
Synchronization, deadlock, wait(), notify(), notifyAll().

• **Day 3: Java I/O Streams:**
Byte Streams: InputStream, OutputStream, FileInputStream, FileOutputStream.
Character Streams: Reader, Writer, FileReader, FileWriter, BufferedReader, BufferedWriter, PrintWriter.

• **Day 4: File Handling & Serialization:**
File class, file operations, serialization, deserialization, transient keyword.

• **Day 5: Java 8 Features:**
Lambda Expressions, Functional Interfaces, Stream API, Optional class, Method Reference, Date & Time API.

o Lab: Build File Management Application.

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WEEK 7: DATABASE & WEB TECHNOLOGIES

• **Day 1: DBMS Basics & SQL Introduction:**
DBMS, RDBMS, normalization, SQL introduction.

• **Day 2: SQL Queries:**
SELECT, INSERT, UPDATE, DELETE, clauses, joins, subqueries, indexes, views.

• **Day 3: JDBC:**
JDBC architecture, JDBC drivers, connectivity, Statement, Prepared Statement, Callable Statement, Result Set.

• **Day 4: CRUD Operations & Transactions:**
Perform CRUD operations, batch processing, transaction management, commit & rollback.

• **Day 5: Servlets & JSP Basics:**
Servlet lifecycle, doGet(), doPost(), request/response objects, session tracking, cookies.
JSP basics, directives, scripting elements, implicit objects, MVC architecture.

o **Lab:** Build JDBC-based Employee Database System.

WEEK 8: HIBERNATE, SPRING, SPRING BOOT & INTERVIEW PREPARATION

• **Day 1: Hibernate Fundamentals:**
Introduction to ORM (Object Relational Mapping), advantages of Hibernate over JDBC, Hibernate architecture, Hibernate lifecycle, Hibernate configuration file (hibernate.cfg.xml), dependencies setup.

• **Day 2: Hibernate Mapping:**
XML Mapping and Annotation Mapping
Annotations: @Entity, @Table, @Id, @GeneratedValue, @Column, @Transient, @Temporal
Primary Key Generation Strategies
Hibernate SessionFactory, Session, Transaction objects.

• **Day 3: Hibernate CRUD Operations:**
Insert, Update, Delete, Select operations using Hibernate.
Methods: save(), persist(), update(), save Or Update(), delete(), get(), load().

• **Day 4: Hibernate Advanced Concepts:**
HQL (Hibernate Query Language)
Criteria API
Pagination in Hibernate
Named Queries
Native SQL Queries.

• **Day 5: Hibernate Relationships Mapping:**
One-to-One Mapping, One-to-Many Mapping, Many-to-One Mapping, Many-to-Many Mapping

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• **Day 6: Spring Framework Basics:**
Introduction to Spring Framework
Modules of Spring
IOC (Inversion of Control)
Dependency Injection (Constructor Injection / Setter Injection)
Bean lifecycle
Spring configuration (XML & Annotations)

• **Day 7: Spring Boot Basics:**
Introduction to Spring Boot
Advantages of Spring Boot
Project Structure
Starter Dependencies
application.properties / application.yml
Spring Boot annotations: @SpringBootApplication, @Component, @Service, @Repository, @Controller, @RestController

• **Day 8: Spring Boot + Database Integration:**
Spring Data JPA
Repository Interface
JpaRepository / CrudRepository
Database configuration in Spring Boot
CRUD operations with Spring Boot + MySQL.

• **Day 9: REST API & Microservices:**
REST API development (GET, POST, PUT, DELETE), Postman testing, microservices basics, system design introduction.

• **Day 10: Interview Preparation:**
Core Java interview questions, Advanced Java questions, JDBC, Hibernate, Servlet/JSP, Spring Boot questions, coding programs.

• **Mock Interview & Resume Preparation:**
HR questions, technical mock interview, resume building, project explanation.

o **Final Project:** Build full-stack **Java Web Application** using Java + MySQL + Hibernate + Spring Boot.

This syllabus is perfect for becoming a Java Developer, Backend Developer, or Full Stack Developer.

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